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DATA DESCRIPTOR

BDSet: A First-ever BDS Satellite-based PPP Augmentation Dataset Since 2023

Chenhao Ouyang^{1,2}, Junbo Shi¹, Cheng Hou¹, Wenjie Peng¹, Jiming Guo¹ & Yibin Yao¹

The BDS precise point positioning (PPP) service broadcasts state space representation (SSR) corrections via PPP-B2b augmentation signals. To maximize research potentials of the service, this study develops a first-ever BDS PPP-B2b augmentation dataset since January 2023, named BDSet. This dataset originates from PPP-B2b augmentation signals collected in Wuhan, China, and covers the processing flow of 'Raw data → Standardized data → Decoded data → Final products', i.e. raw data from receiving chip, standardized PPP-B2b augmentation messages, decoded PPP-B2b SSR corrections, and recovered PPP-B2b precise products which can be directly utilized in positioning, navigation, and timing (PNT) applications. The developed BDSet has been continuously updating since January 1, 2023. The dataset in 2023, 2024 and 2025 occupies a storage memory of 142.7 GB, containing 255,405,156 PPP-B2b augmentation messages. Message amounts for BDS C59, C60, C61 and C62 satellites are 87,080,707, 80,148,878, 13,107,885 and 75,067,686, respectively. The developed BDSet provides a substantial data base for applications in GNSS PNT, geoscience, meteorology and so on. Meanwhile, the three-year dataset can better facilitate artificial intelligence (AI)-empowered BDS big data mining in the future.

Background & Summary

Satellite-based precise point positioning (PPP) services enable high-accuracy positioning for massive users by broadcasting state space representation (SSR) corrections via satellite communication links. Due to the property of satellite communication links, PPP services can be applied in regions with limited or even no terrestrial internet coverage, such as at sea and in less-populated areas. Existing satellite-based PPP services can be categorized into commercial and non-commercial types. Commercial services require user subscription fees, such as Trimble's CenterPoint RTX, Hemisphere's Atlas, NovAtel's TerraStar, VERIPOS's APEX, FUGRO's MarineStar, Hi-Target's Hi-RTP, Qianxun SI's XStar and so on. In contrast, non-commercial services are available for free, such as Centimeter-Level Augmentation Service (CLAS) of QZSS¹, High Accuracy Service (HAS) of Galileo², and PPP service of BDS³.

On December 27, 2019, the first version of BDS PPP-B2b signal interface control document (ICD) was released, marking official launch of the BDS PPP service⁴. This service is designed as a unique component along with regular positioning, navigation, and timing (PNT) services. Specifically, the BDS PPP service generates BDS/GPS satellite orbit, clock offset, and differential code bias (DCB) corrections at intervals of 48/6/48 s, respectively, based on 12 continuous tracking stations located within China⁵. These generated SSR corrections are encoded into PPP-B2b augmentation messages, which are then uploaded to BDS geostationary earth orbit (GEO) satellites. GEO satellites subsequently modulate augmentation messages into PPP-B2b augmentation signals which are broadcast to the earth. Once these augmentation signals were received, users can obtain improved performance over the traditional single point positioning using a standalone receiver⁶.

Currently, several types of receiving chips are reported to support the reception of PPP-B2b augmentation signals, such as ComNav K803, K823, and M300 Pro used by Ouyang *et al.*⁷, Xu *et al.*⁸, and Yang *et al.*⁹, the FRII-Plus used by Xu *et al.*¹⁰, Xu *et al.*¹¹, and Tao *et al.*¹², and the Septentrio Mosaic X5 used by Zhou *et al.*¹³. Additionally, the Redmi K60 Ultra smartphone used by Wang *et al.*¹⁴ has also achieved the PPP-B2b signal reception. Once the receiving chip tracks PPP-B2b augmentation signals, it decodes the message to obtain SSR

¹School of Geodesy and Geomatics, Wuhan University, Wuhan, 430079, China. ²Hubei LuoJia Laboratory, Wuhan, 430079, China. ✉e-mail: jbsi@sgg.whu.edu.cn

corrections, which are then combined with the broadcast ephemeris to recover precise orbits, clock offsets and DCBs¹⁵. The detailed recovery procedure is provided in the official BDS PPP-B2b ICD on pages 33 and 34⁴.

Based on the recovered PPP-B2b precise products, numerous studies have been conducted, focusing on product accuracy evaluation, positioning algorithm development, as well as zenith tropospheric delay (ZTD) and precipitable water vapor (PWV) estimation. Regarding the product accuracy evaluation, existing studies have used various post-processed precise products as references to evaluate PPP-B2b precise products. Taking the final products from Wuhan University as references, Tao *et al.*¹⁶ evaluated one-week PPP-B2b precise products in August 2020, obtaining orbit root mean square errors (RMSEs) of 0.065/0.240/0.210 m and 0.062/0.345/0.213 m in radial (R)/along-track (A)/cross-track (C) directions for BDS and GPS satellites, respectively. The standard deviations (STDs) of clock offsets were 0.11 ns and 0.13 ns, respectively. Using final products from the International GNSS Monitoring and Assessment System (iGMAS) as references, Xu *et al.*¹⁷ evaluated one-month PPP-B2b precise product during October 2020, yielding orbit RMSEs of 0.068/0.334/0.366 m and 0.156/0.398/0.518 m in R/A/C directions for BDS medium earth orbit (MEO) and inclined geosynchronous orbit (IGSO) satellites, respectively, and a clock offset STD of 0.2 ns. Referencing rapid products from the German Research Centre for Geosciences (GFZ), Ren *et al.*¹⁸ evaluated one-week PPP-B2b precise products in August 2020, reporting R/A/C orbit RMSEs of 0.072/0.248/0.259 m and 0.073/0.310/0.219 m, and clock offset STDs of 0.05–0.18 ns and 0.05–0.15 ns for BDS and GPS satellites, respectively. Meanwhile, differences between the PPP-B2b DCB and International GNSS Service (IGS) DCB products were reported to be less than 2 ns. Based on one-year PPP-B2b precise products from May 1, 2021 to May 1, 2022, Sun *et al.*¹⁹ assessed the long-term performance of BDS PPP service, obtaining R/A/C orbit RMSEs of 0.078/0.325/0.356 m and 0.086/0.459/0.362 m for BDS and GPS, respectively, and clock offset STDs of 0.16 ns and 0.20 ns. It should be noted that all previous studies have proven that the improvement of PPP-B2b precise products over broadcast ephemeris.

In terms of general positioning algorithms using PPP-B2b precise products, existing researches mainly focus on optimizing functional and stochastic models for open sky and obstructed environments. In open sky environments, Xu *et al.*¹⁰ proposed an improved functional model that treated signal in space range errors as unknown parameters. This model reduced the impact of PPP-B2b orbit/clock offset/DCB errors on coordinate parameter estimation, resulting in an average positioning accuracy improvement of 21.1%/25.4% in horizontal/vertical directions for vehicular and maritime kinematic positioning. Dai *et al.*²⁰ introduced an enhanced stochastic model based on multipath errors, which shortened the PPP convergence time by 10%/14%/9% in East (E)/North (N)/Up (U) directions, achieving E/N/U accuracies of 9.9/6.5/14.0 cm based on PPP-B2b precise products. Tao *et al.*¹² realized integer ambiguity fixing by estimating phase observable-specific biases (OSB) based on PPP-B2b precise products, reducing PPP convergence time from 26.1/14.1 minutes to 3.4/4.3 minutes in horizontal/vertical directions, and improving positioning accuracy by 68.2%. To maintain positioning continuity during correction interruptions, Shi *et al.*²¹ trained a neural network for PPP-B2b clock offset prediction. Using predicted versus real clock offsets results in less than 2.6/4.9 cm positioning accuracy difference in horizontal/vertical directions after convergence. In obstructed environments, existing studies mainly focused on optimizing functional models formed by the combination of GNSS and other sensors. Xu *et al.*²² developed a GNSS/inertial navigation system (INS) integrated system based on PPP-B2b precise products. In vehicular experiments under urban environments, accuracies of GNSS-only positioning were 1.14/2.00/2.44 m in E/N/U directions, which were improved by 64.1%/77.8%/73.8% when GNSS/INS integration was used. Shang *et al.*²³ developed a GNSS/INS/camera integrated system based on PPP-B2b precise products, achieving lane-level positioning with smartphones. In urban environments, positioning accuracy was improved from 1.48 m to 0.99 m. Guo *et al.*²⁴ proposed a GNSS/INS/light detection and ranging (LiDAR) integrated system based on PPP-B2b precise products, improving positioning accuracy from meter- to sub-meter-level in areas surrounded by buildings and trees.

Previous studies have demonstrated the applicability of PPP in meteorology^{25,26}. Using PPP-B2b precise products, several researches have evaluated the accuracy of PPP-derived ZTD and PWV. For example, Wang *et al.*²⁷ derived ZTD for 44 IGS stations from September 7 to 13, 2022, based on PPP-B2b precise products. The results showed ZTD RMSEs ranging from 10–20 mm for stations in China and 15–30 mm for stations in Europe. Zhou *et al.*¹³ derived ZTD based on PPP-B2b precise products for 15 IGS MEGX stations between June 30 and August 17, 2023. Compared with ZTD generated from WUM post-processed precise products, IGS final ZTD products, radio-sonde data, and European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ERA5) data, the RMSEs were 19.2 mm, 21.5 mm, 25.7 mm, and 28.2 mm, respectively. Cao *et al.*²⁸ assessed the PWV derived from PPP-B2b precise products at a station in Beijing, China, reporting a RMSE of 1.8 mm during January 2023. Xu *et al.*¹¹ generated PWV for 17 coastal stations in China between June 6 and 10, 2022, using PPP-B2b precise products, reporting mean absolute errors of 2.10 mm, 2.60 mm, and 2.48 mm for stations in the North, South and East China Sea regions, respectively.

Although the BDS PPP service has been explored from the aforementioned aspects, in-depth studies are still hindered by the following issues. First, the format of PPP-B2b augmentation data are not unified for various receiving chips. Second, processed PPP-B2b precise products are directly used, without investigating error patterns of underlying raw augmentation data. Third, the longest duration of PPP-B2b augmentation data in previous studies was one year, which cannot timely reflect the spatiotemporal variability of BDS PPP service during different operational phases. Fourth, algorithmic researches primarily focus on limited scenarios, lacking validation of generalization ability across diverse scenarios such as deformation monitoring²⁹, marine vessel positioning³⁰ and so on.

To address these issues, this study designs and implements a first-ever BDS PPP-B2b augmentation dataset, named BDS_{Set}, which covers the full data processing flow of ‘Raw data → Standardized data → Decoded data → Final products’, i.e. raw messages output by receiving chips, standardized PPP-B2b augmentation messages following the BDS PPP-B2b signal ICD, decoded PPP-B2b SSR corrections, and recovered PPP-B2b orbit/clock offset/DCB products based on SSR corrections and broadcast ephemerides. In the following section, the

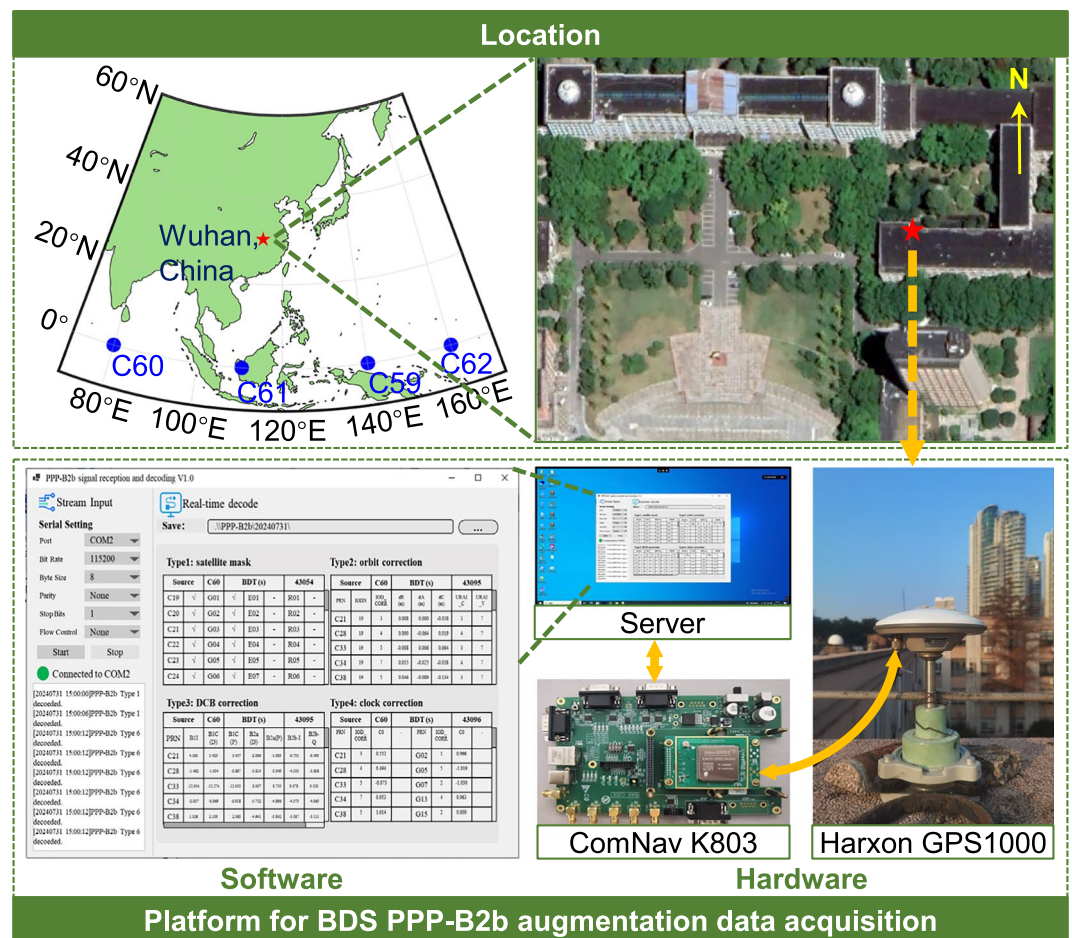


Fig. 1 Configuration for the PPP-B2b augmentation data acquisition.

developed platform for PPP-B2b augmentation data acquisition is first introduced. Next, the dataset design is explained, detailing the data processing workflow and data classification scheme. Then, a three-year dataset from Jan. 2023 to Dec. 2025 is developed with implementation details including naming conventions, storage paths, and formats of various files. Finally, the developed dataset is validated from several aspects, such as data size, message amount, spatiotemporal patterns of PPP-B2b SSR corrections, and accuracies of recovered PPP-B2b precise products.

Methods

This section first introduces the developed platform for PPP-B2b augmentation data acquisition. Then, procedures used to create the dataset are explained.

Dataset acquisition. Figure 1 illustrates the environment and platform for PPP-B2b augmentation data collection. The collection site locates on the rooftop of a building in the city of Wuhan, China, providing an open sky environment. BDS GEO satellites broadcasting the PPP-B2b augmentation signals are C59, C60, C61, and C62, positioned above the equator at longitudes of 140°, 80°, 110.5°, and 160°E, respectively (<https://www.bdsupport.cn/status/constellation>). The data acquisition platform consists of both hardware and software components. The hardware includes a Harxon GPS1000 antenna, a ComNav K803 receiving chip with firmware version 620F1.23924.1, and a data processing server. The antenna is connected to the receiving chip via cables, and the receiving chip communicates with the server through an RS232 serial port. The software deployed on the server collects PPP-B2b augmentation messages and broadcast ephemerides.

Dataset design. Figure 2 illustrates the classification of PPP-B2b augmentation dataset. Generating PPP-B2b precise products from raw data requires four stages, each correlating to a specific type of PPP-B2b augmentation dataset. Therefore, this study classifies the PPP-B2b augmentation dataset into four levels: Levels 0–3 correspond to raw messages from the receiving chip, standardized PPP-B2b augmentation messages following the ICD, decoded PPP-B2b SSR corrections, and recovered PPP-B2b orbit/clock offset/DCB products, respectively.

Level 0: Raw messages from receiving chip. The receiving chip outputs binary messages containing both the PPP-B2b augmentation messages and BDS/GPS broadcast ephemerides in a chip-specific format.

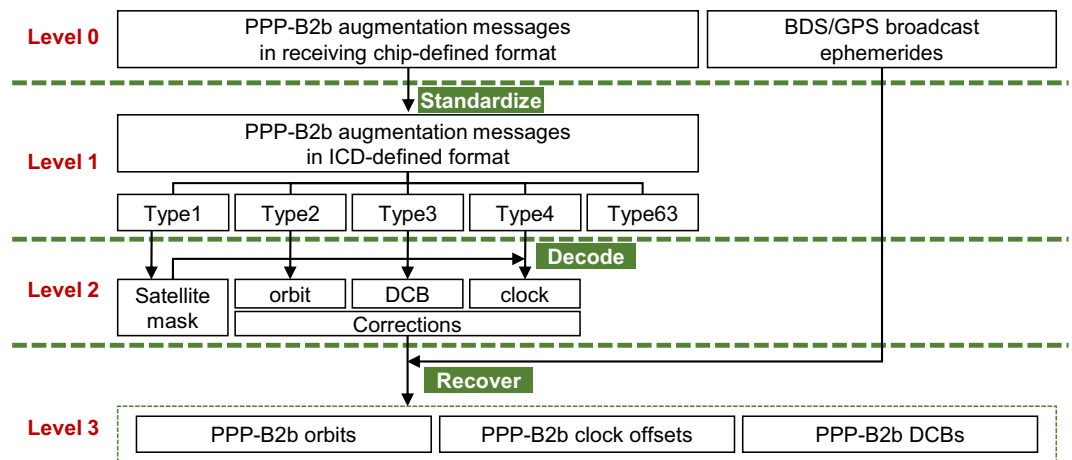


Fig. 2 Classification of PPP-B2b augmentation dataset.

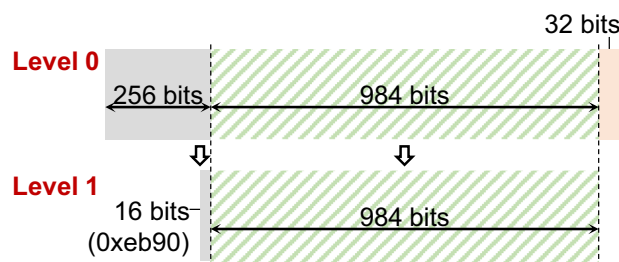


Fig. 3 Structure of PPP-B2b augmentation messages in Level 0 and Level 1. The green parts of the two messages are identical.

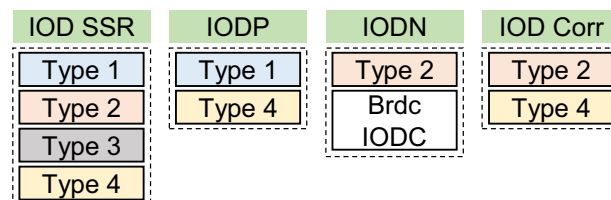


Fig. 4 Relationships among various IODs in PPP-B2b augmentation messages.

Level 1: Standardized PPP-B2b augmentation messages. Augmentation messages in Level 0 are not organized in the ICD-defined format. To standardize these messages, PPP-B2b augmentation messages are first extracted from raw messages. Then, they are re-encoded according to the format defined in the BDS PPP-B2b signal ICD.

To better understand the re-encoding process, this study takes ComNav K803 receiving chip-defined format as the example and compare PPP-B2b augmentation message structures of Level 0 and Level 1 in Fig. 3. Both messages contain identical 984 bits, corresponding to bits 257–1,240 in Level 0 and bits 17–1,000 in Level 1. The re-encoding process involves removing the first 256 bits and the last 32 bits of PPP-B2b augmentation message in Level 0, and adding a 16-bit preamble (0xeb90) to the beginning. The standardized PPP-B2b augmentation messages in Level 1 contain five types: Type 1 (satellite mask), Type 2 (orbit corrections), Type 3 (DCB corrections), Type 4 (clock offset corrections) and Type 63 (null message).

Level 2: Decoded PPP-B2b SSR corrections. Type 1/2/3/4 messages in Level 1 are decoded to obtain PPP-B2b SSR corrections in American Standard Code for Information Interchange (ASCII). Types 2 and 3 include satellite Pseudo-Random Noise (PRN) codes, enabling direct association of satellites with their orbit/DCB corrections. While Type 4 does not contain satellite PRN codes, satellite mask in Type 1 is required to associate satellites with their clock offset corrections.

Level 3: Recovered PPP-B2b orbit/clock offset/DCB products. By combining broadcast ephemerides from Level 0 and PPP-B2b SSR corrections from Level 2, and performing multiple Issue of Data (IOD) matchings¹⁵, PPP-B2b orbit/clock offset/DCB products are recovered and can be directly used for subsequent PNT applications.

Field	Length	Content
PPP-B2b	7	Denotes data from PPP-B2b augmentation signals
YYYYDOY (Date)	7	'YYYY': year 'DOY': day of year
CCC (Content)	3	'MIX': mixed messages from receiving chip 'B2B': PPP-B2b augmentation messages 'SSR': PPP-B2b SSR corrections 'ORB': PPP-B2b orbit 'CLK': PPP-B2b clock offset 'DCB': PPP-B2b DCB
FFF (Format)	3	'RAW': binary format defined by receiving chip 'B2B': binary format defined by PPP-B2b signal ICD ³² 'SSR': ASCII PPP-B2b SSR correction format defined by this study 'SP3': SP3-d format defined by IGS ³⁴ 'CLK': Clock RINEX 3.04 format defined by IGS ³⁵ 'BSX': Bias-SINEX V1.00 format defined by IGS ³⁶

Table 1. File Naming Convention in The BDS PPP-B2b Augmentation Dataset.

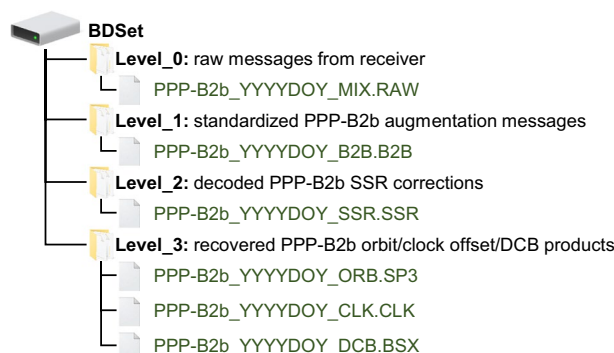


Fig. 5 File directory structure of the developed BDS PPP-B2b augmentation dataset.

The IOD matching serves to associate SSR corrections with broadcast ephemerides³¹. PPP-B2b SSR corrections contain four types of IOD: IOD SSR, IOD PRN mask (IODP), IOD Navigation (IODN), and IOD Correction (IOD Corr). Figure 4 illustrates relationships among these four IODs: IOD SSR associates Type 1 with Type 2/3/4, IODP associates Type 1 with Type 4, IODN associates Type 2 with broadcast ephemerides, and IOD Corr associates Type 2 with Type 4. With all IODs matched, the broadcast orbit/clock offset are first calculated, followed by the PPP-B2b orbit/clock offset calculation³².

Data Record

The implemented dataset is currently available at <https://doi.org/10.57760/sciencedb.cast.0000933>. Based on the dataset design introduced in the previous section, this section presents the dataset implementation, i.e. storing the four levels of PPP-B2b augmentation data as files. Firstly, the file naming conventions and storage paths are introduced. Then, file formats in various storage path are described in detail.

File naming convention and structure. Table 1 tabulates the file naming convention for the developed BDS PPP-B2b augmentation dataset. All filenames are fixed at 23 characters in length and follow the format 'PPP-B2b_YYYYDOY_CCC.FFF', where 'PPP-B2b' indicates that the data originates from BDS PPP-B2b augmentation signals; 'YYYYDOY' refers to data acquisition date; 'CCC' represents the file content; and 'FFF' denotes the file format.

Figure 5 illustrates the file directory structure of BDS PPP-B2b augmentation dataset. The root folder is named as BDSet, which is the name of the developed dataset. Four sub-folders are set up according to the data generation sequence: Level_0, Level_1, Level_2, and Level_3, with daily files stored in each sub-folder. The file contents, along with their filenames, are presented below:

Files in Level_0 contain raw messages from the receiving chip, with filenames as 'PPP-B2b_YYYYDOY_MIX.RAW'.

Files in Level_1 contain standardized PPP-B2b augmentation messages in the ICD-defined format, with filenames as 'PPP-B2b_YYYYDOY_B2B.B2B'.

Files in Level_2 contain decoded PPP-B2b SSR corrections, with filenames as 'PPP-B2b_YYYYDOY_SSR.SSR'.

Files in Level_3 contain recovered PPP-B2b orbit/clock offset/DCB products, with filenames as 'PPP-B2b_YYYYDOY_ORB.SP3'/'PPP-B2b_YYYYDOY_CLK.CLK'/'PPP-B2b_YYYYDOY_DCB.BSX', respectively.

For better understanding, Table 2 summarizes filenames, file contents, and file formats in each sub-folder. Files in sub-folder Level_0 and Level_1 are in binary format, while files in Level_2 and Level_3 are text files encoded in ASCII. Following sections provide detailed descriptions of file formats.

Sub-folder	File name	File content	File format
Level_0	PPP-B2b_YYYYDOY_MIX.RAW	Raw messages from receiving chip	Binary format defined by receiving chip
Level_1	PPP-B2b_YYYYDOY_B2B.B2B	Standardized PPP-B2b augmentation messages	Binary format defined by PPP-B2b signal ICD
Level_2	PPP-B2b_YYYYDOY_SSR.SSR	Decoded PPP-B2b SSR corrections	ASCII PPP-B2b SSR correction format defined by this study
Level_3	PPP-B2b_YYYYDOY_ORB.SP3/PPP-B2b_YYYYDOY_CLK.CLK/PPP-B2b_YYYYDOY_DCB.BSX	Recovered PPP-B2b orbit/clock offset/DCB products	ASCII SP3/CLK/BSX format defined by IGS

Table 2. Filenames, File Contents and File Formats of Various Files in Each Sub-folder.

Field #	Content	Length (bit)
1	Preamble, fixed to '0xaa4412' in hexadecimal format	24
2	Header length in byte	8
3	Message Type	16
4	Reserved	16
5	Message length in byte	16
6	Reserved	32
7	GPS Week	16
8	Seconds within the week in ms	32
9	Reserved	48
10	Receiver software version	16
11	Message content	(Field #5) × 8
12	Cyclic redundancy check (CRC) code*	32

Table 3. Message Structure Defined by ComNav K803 Receiving Chip. *The CRC code is a 32-bit checksum based on Field #1–11 with the purpose of accidental error detection during transmission. More details can refer to³⁹.

Level 0: raw messages from receiving chip. Table 3 lists the structure of raw messages in the sub-folder Level_0. There are 12 fields in each raw message. The first 24 bits serve as preamble, represented in hexadecimal format as '0xaa4412'. Bits 33–48 indicate the message type. Type 1697 corresponds to PPP-B2b augmentation message. Type 72 corresponds to BDS-3 broadcast ephemeris, and Type 71 corresponds to GPS broadcast ephemeris. It should be noted that bytes in each field are arranged in big-endian order, meaning that the byte order must be reversed before converting to decimal values.

Take one Type 1697, one Type 72, and one Type 71 message as examples, Fig. 6 shows their contents in hexadecimal format. It can be observed that three messages have the same preamble (0xaa4412), with message lengths of 1272 bits = 159 bytes × 8, 1984 bits = 248 bytes × 8, and 2048 bits = 256 bytes × 8, respectively. Values of Field #3 for the three messages are 0xa106, 0 × 4800, and 0 × 4700, which are 1697, 72, and 71 in decimal format, respectively, i.e. Type 1697, 72 and 71.

Level 1: standardized PPP-B2b augmentation messages. Table 4 presents the structure of PPP-B2b augmentation messages in the sub-folder Level_1. The total length of each message is fixed at 1000 bits, with the first 16 bits serving as preamble, represented in hexadecimal format as '0xeb90'. Bits 17–22 indicate the PRN code of GEO satellite broadcasting the message. Bits 23–28 represent the PPP service status, where 0 and 1 denote service available and unavailable, respectively. Bits 29–1000 contain the message content, with the first six bits indicating message type. Field #2–4 correspond to the green part in Fig. 3.

Figure 7 displays five consecutive PPP-B2b augmentation messages broadcast by C59 in hexadecimal format, corresponding to Type 1, 2, 3, 4, and 63, respectively. It can be seen that all five messages have the same preamble (0xeb90), and each has a length of 1000 bits = 125 bytes × 8. By converting the 3rd to 5th bytes of each message to binary format, the GEO satellite PRN code and the message type can be determined, which correspond to Field #2 and #4 in Table 4, respectively. As shown in the figure, the GEO satellite PRN codes for five messages are all '111011', which corresponds to 59 in decimal format, i.e. BDS GEO satellite C59. Message types for five messages are '000001', '000010', '000011', '000100', and '111111', which correspond to 1, 2, 3, 4, and 63 in decimal format, i.e. Type 1, 2, 3, 4 and 63.

Level 2: decoded PPP-B2b SSR corrections. Table 5 presents the format of PPP-B2b SSR corrections in sub-folder Level_2, which follows an ASCII structure defined in this study. Each correction consists of two parts: the Header and the Body.

Header: each Type 1, 2, 3, and 4 occupies a single line with lengths of 97, 79, 79, and 99 characters, respectively. The first three characters represent the preamble, which are '>SAT', '>ORB', '>DCB', and '>CLK', respectively. The header contains two time-related fields: the message reception time and the correction reference

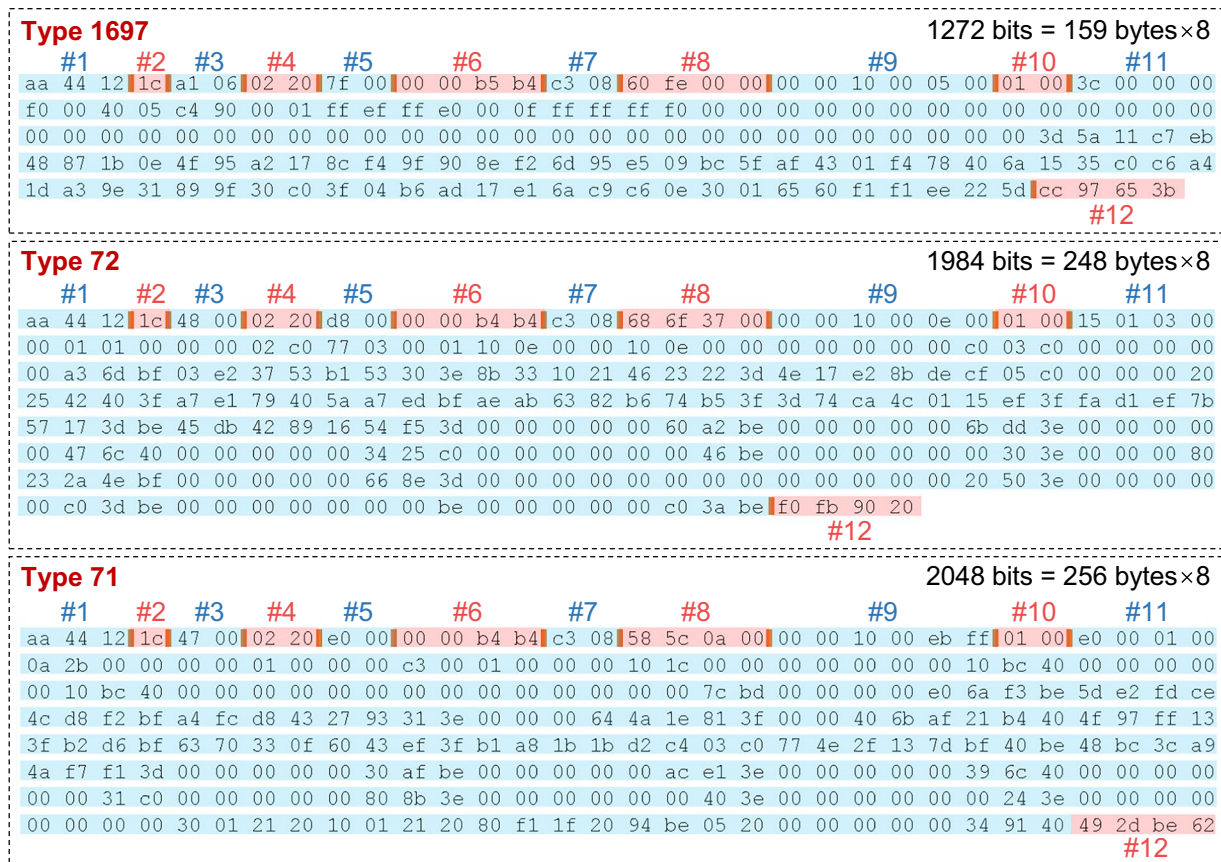


Fig. 6 Example of Level_0 file content in hexadecimal format with 12 fields separated by orange lines. Field #3 denotes message type, i.e. 0xa106, 0 × 4800 and 0 × 4700 denote Type 1697, 72 and 71, respectively.

Field #	Content	Length (bit)
1	Preamble, fixed to '0xeb90' in hexadecimal format	16
2	GEO satellite PRN	6
3	PPP service status (0/1 for available/unavailable)	6
4	Message content, the first six bits denote message type	972

Table 4. Message Structure Defined by BDS PPP-B2b Signal ICD³².

time. These two fields enable the calculation of correction reception latency, which will be expanded in the next section.

Body: Type 1 occupies four lines, each recording PRN codes of valid satellites for one GNSS constellation, with each line having a length of (1 + number of satellites × 3). Types 2, 3, and 4 occupy several lines which equal to the number of satellites, with each line recording the orbit, DCB, and clock offset correction for one satellite. The line lengths for the three message types are 93, (3 + number of signals × 16), and 30, respectively.

Taking adjacent Type 1/2/3/4 as examples, Fig. 8 illustrates the PPP-B2b SSR corrections recorded in the sub-folder Level_2. The blue and red blocks in the figure represent the defined Header and Body in Table 5, respectively. For better understanding, most fields display both the field name and the corresponding value, such as 'PPPStatus' followed by '0', indicating that the PPP service is available.

Level 3: recovered PPP-B2b orbit/clock offset/DCB products. Taking PPP-B2b precise products on January 1, 2023 as an example, Fig. 9 shows recovered PPP-B2b orbits, clock offsets, and DCBs in the sub-folder Level_3. File formats for the three products follow IGS-defined SP3, CLK, and BSX formats^{34–36}, respectively. It should be noted: (1) regarding the orbit reference points, PPP-B2b product refers to the satellite antenna phase centers using BDS B3I observation and GPS L1/L2 ionosphere-free (IF) observation, respectively. (2) regarding the clock offset references, PPP-B2b product refers to BDS B3I observation and GPS L1/L2 IF observation, respectively.

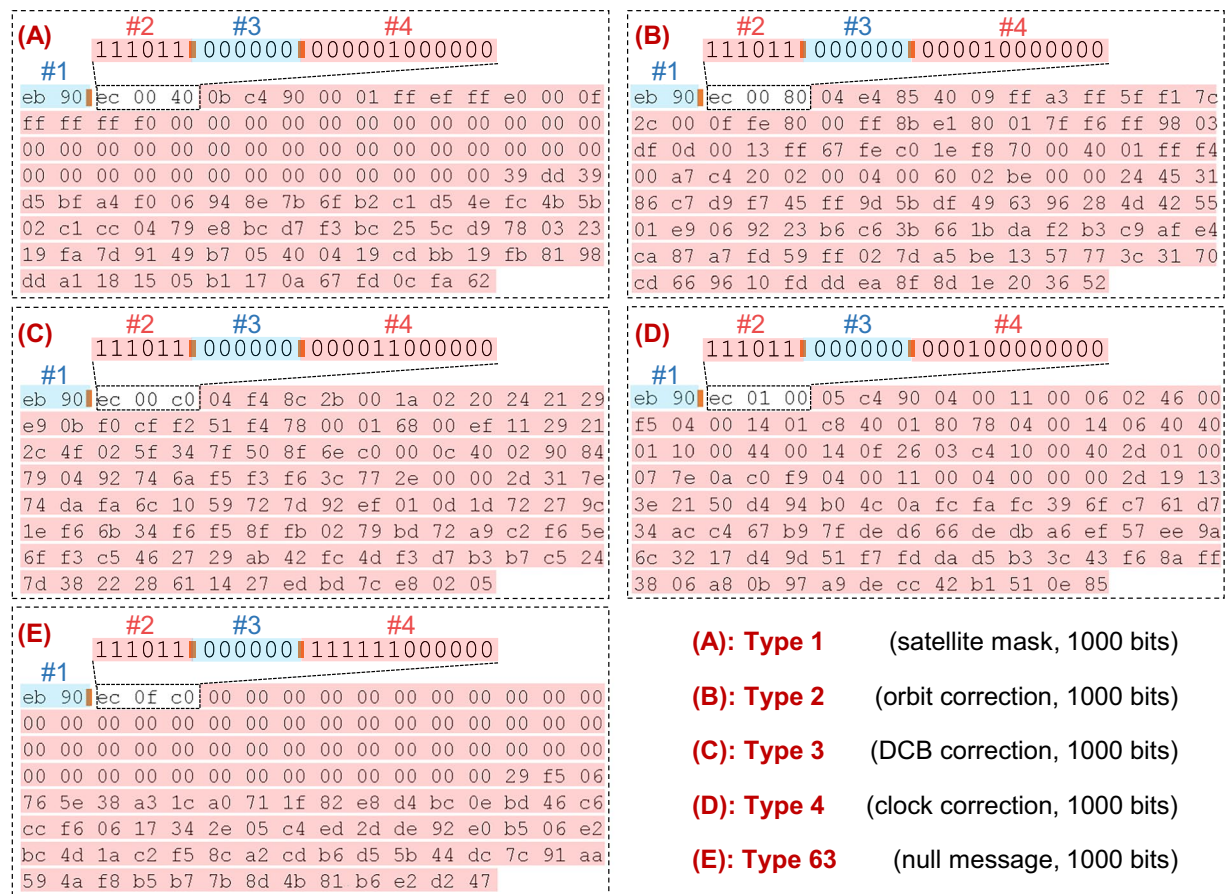


Fig. 7 Example of Level_1 file content in hexadecimal format with four fields separated by orange lines. The first six bits in Field #4 denote message type, i.e. binary ‘000001’, ‘000010’, ‘000011’, ‘000100’ and ‘111111’ denote Type 1, 2, 3, 4 and 63, respectively.

	Type	Content	Length
Header	1–4	1) Preamble	4
		2) Message reception time in GPS Time (GPST)	24
		3) GEO satellite PRN code	4
		4) Status of service	12
		5) Correction reference time, second of day (SoD) in BeiDou Time (BDT)	15
		6) IOD SSR	10
		7) IODP (only for Type 1 and 4)	8
		8) # of valid satellites (only for Type 1)	20
		9) SubType1 (only for Type 4)	12
		10) # of corrections (only for Type 2, 3 and 4)	10
Body	1	1) PRN codes of valid satellites	1 + sat# × 3
	2	1) Satellite PRN code 2) IODN and IOD Corr 3) Orbit corrections, unit: m 4) User Range Accuracy Index (URAI)	3 26 36 28
	3	1) Satellite PRN code 2) DCB correction, unit: m	3 sig# × 16
	4	1) Satellite PRN code 2) IOD Corr 3) Clock correction, unit: m	3 15 12

Table 5. ASCII PPP-B2b SSR Correction Format Defined in This Study.

Technical Validation

Based on the dataset implementation described in the previous section, this study has developed the BDS PPP-B2b augmentation dataset, i.e. the BDSset. The dataset started since January 1, 2023, and will be updated on a monthly basis. For the year of 2023, 2024 and 2025, the dataset occupies a storage memory of 142.7 GB, containing 255,405,156 PPP-B2b augmentation messages. Message amounts for BDS C59, C60, C61 and C62

Header	Body
Type 1 (satellite mask)	
<pre>>SAT 2023 01 01 01 00 17.12 C59 PPPStatus 0 BDT_SoD 3598 IOD_SSR 1 IODP 2 C G E R 27 32 0 0 C 19 20 21 22 23 24 25 26 27 28 29 30 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 G 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 E R</pre>	
Type 2 (orbit correction)	
<pre>>ORB 2023 01 01 01 00 28.12 C59 PPPStatus 0 BDT_SoD 3591 IOD_SSR 1 SatNum 6 C21 IODN IOD_Corr 0 5 dR dA dC -0.0448 0.0000 -0.0896 URAI_Class URAI_Value 3 7 C24 IODN IOD_Corr 0 3 dR dA dC -0.0544 -0.0384 0.0640 URAI_Class URAI_Value 3 7</pre>	
Type 3 (DCB correction)	
<pre>>DCB 2023 01 01 01 01 9.12 C59 PPPStatus 0 BDT_SoD 3639 IOD_SSR 1 SatNum 3 C21 B1I 3.5360 B1C(D) 4.3690 B1C(P) 4.5050 B2a(D) -3.1280 B2a(P) -2.0740..... C24 B1I 1.3940 B1C(D) 2.4310 B1C(P) 2.4820 B2a(D) -6.3070 B2a(P) -5.4740..... C26 B1I -1.3260 B1C(D) -0.1360 B1C(P) 0.0000 B2a(D) -5.8480 B2a(P) -4.9980.....</pre>	
Type 4 (clock correction)	
<pre>>CLK 2023 01 01 01 01 12.12 C59 PPPStatus 0 BDT_SoD 3646 IOD_SSR 1 IODP 2 SubType1 0 SatNum 9 C21 IOD_Corr 6 C0 0.6640 C24 IOD_Corr 4 C0 -0.2464</pre>	

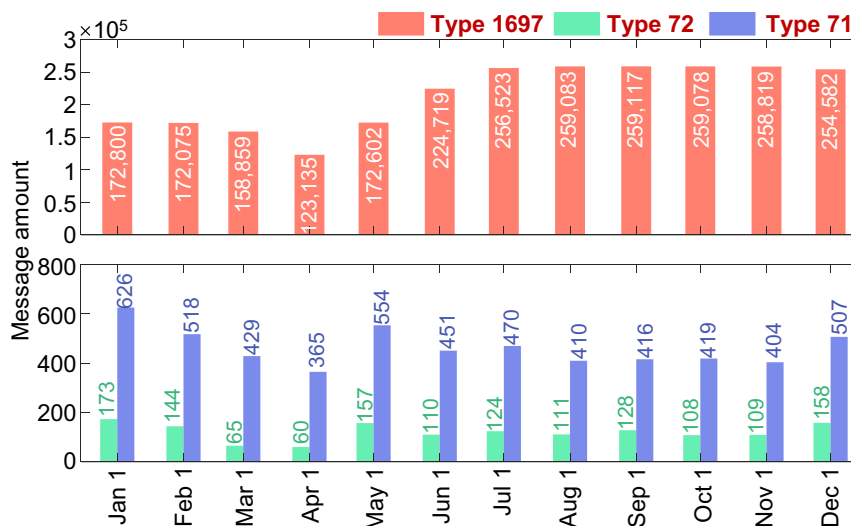
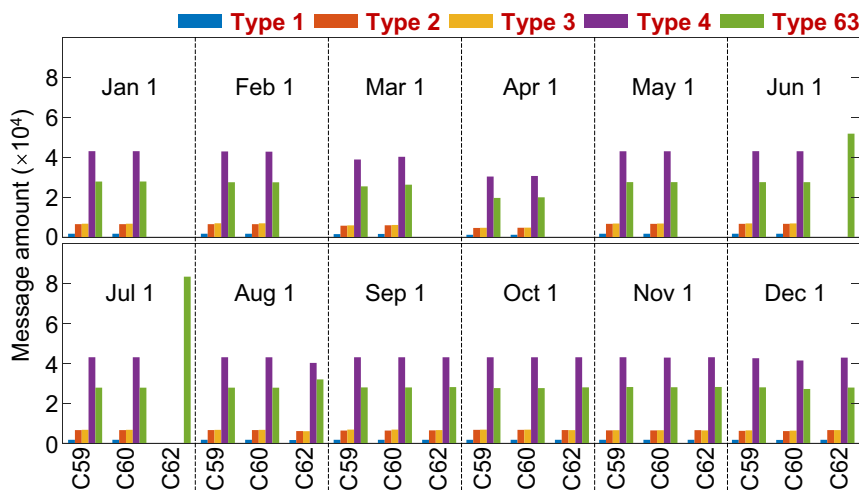
Fig. 8 Example of Level_2 file content. The blue and red region represent Header and Body of PPP-B2b SSR corrections, respectively.

SP3 file (PPP-B2b orbit)	
<pre>#dP2023 01 01 0 0 0.00000000 BDCS BDS /* BDS: CNAV1 + SSR, B3I APC /* GPS: LNAV + SSR, L1/L2 IF APC * 2023 1 1 0 0 5.00000000 PC21 -16143.095871 11938.362649 19400.853433 999999.999999 PC22 -27439.096917 1116.461362 5013.317273 999999.999999 PG08 20843.671227 1752.353003 16540.474356 999999.999999 PG10 5111.459156 14036.023880 22126.127220 999999.999999</pre>	
CLK file (PPP-B2b clock)	
<pre>3.04 C RINEX VERSION / TYPE BDS: CNAV1 + SSR, B3I COMMENT GPS: LNAV + SSR, L1/L2 IF COMMENT END OF HEADER AS C21 2023 1 1 0 0 0.000000 1 -9.205655769332e-04 AS C22 2023 1 1 0 0 0.000000 1 -7.097266340954e-04 AS G08 2023 1 1 0 0 0.000000 1 -1.013905042845e-04 AS G10 2023 1 1 0 0 0.000000 1 -1.850906478535e-05</pre>	
BSX file (PPP-B2b DCB)	
<pre>%=BIA 1.00 WHU 00:000:00000 BDS 2023:001:00000 2023:002:00000 A 00000000 +BIAS/SOLUTION *BIAS SVN_ PRN STATION_ OBS1 OBS2 BIAS_START_ BIAS_END_ UNIT _ESTIMATED_VALUE_ _STD_DEV_ OSB C21 C2I 2023:001:00000 2023:002:00000 ns 11.7948 0.0000 OSB C21 C6I 2023:001:00000 2023:002:00000 ns 0.0000 0.0000</pre>	

Fig. 9 Example of Level_3 file content. The SP3 file follows SP3-d format, the CLK file follows Clock RINEX 3.04 format, and the BSX file follows Bias-SINEX V1.00 format.

satellites are 87,080,707, 80,148,878, 13,107,885 and 75,067,686, respectively. Taking data in 2023 as an example, this section provides an initial evaluation of the dataset.

Sub-folder	File amount	File size (GB)
Level_0	365	11.62
Level_1	365	9.11
Level_2	365	16.64
Level_3	1095	6.24
Total	2190	43.61

Table 6. File Amount and Size for BDSet During 2023.**Fig. 10** Type 1697/72/71 amount for the first day of each month in 2023.**Fig. 11** Type 1/2/3/4/63 amount for the first day of each month in 2023.

File size. Table 6 lists the amount of files and the corresponding storage memory of the dataset in 2023. The number of files in the sub-folder Level_0-3 are 365, 365, 365, and 1095, with memory sizes of 11.62 GB, 9.11 GB, 16.64 GB, and 6.24 GB, respectively, totaling 43.61 GB. Taking files on January 1, 2023 as examples, the file 'PPP-B2b_2023001_MIX.RAW' in Level_0 occupies 26.3 MB; 'PPP-B2b_2023001_B2B.B2B' in Level_1 occupies 20.5 MB; 'PPP-B2b_2023001_SSR.SSR' in Level_2 occupies 39.6 MB; 'PPP-B2b_2023001_ORB.SP3', 'PPP-B2b_2023001_CLK.CLK', and 'PPP-B2b_2023001_DCB.BSX' in Level_3 occupy 2.1 MB, 16.0 MB, and 0.02 MB, respectively.

Evaluation of Level 0. Decoding raw messages in the sub-folder Level_0 reveals that a total of 78,450,383 messages were collected in 2023, with the amounts of Type 1697, 72, and 71 being 78,246,531, 42,711, and 161,141, respectively. Taking the first day of each month as examples, Fig. 10 illustrates the variation of message amount over time. It can be observed that:

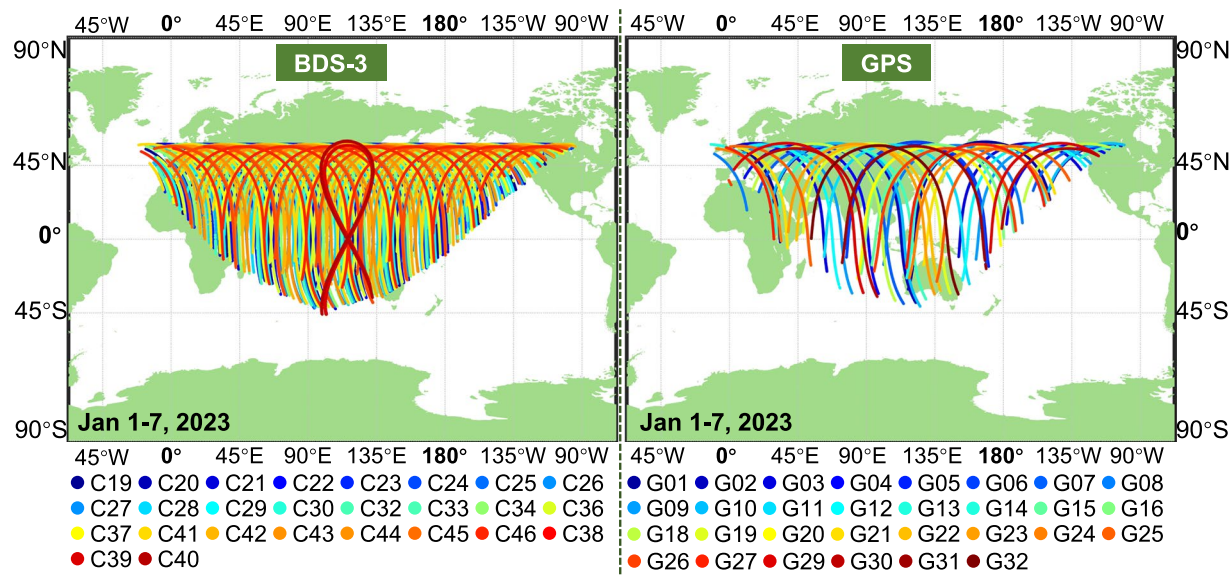


Fig. 12 Ground tracks for BDS (left) and GPS (right) satellites with valid PPP-B2b SSR corrections during Jan 1-7, 2023.

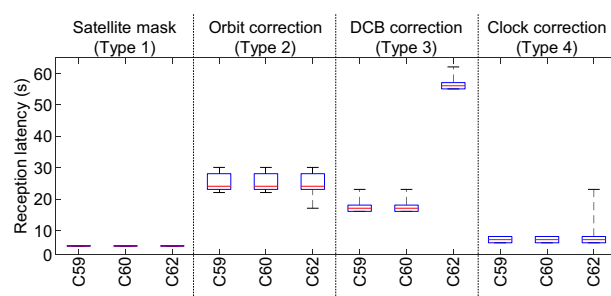


Fig. 13 Reception latencies for satellite masks, orbit corrections, DCB corrections and clock corrections during Dec 1, 2023.

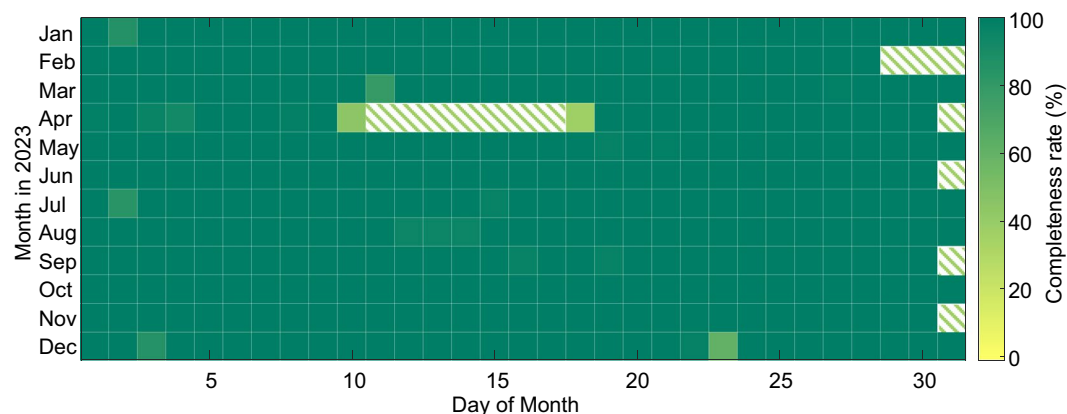


Fig. 14 Daily completeness rate of PPP-B2b precise products in 2023.

- (1) For Type 1697, the number of messages from January to May ranges between 123,135 and 172,800, and then increases since June, ranging between 224,719 and 259,117. The increase in June is due to the launch of the 4th BDS-3 GEO satellite C62 on May 17, 2023, which started broadcasting PPP-B2b augmentation messages since May 24, 2023.
- (2) For Type 72 and 71, due to different updating intervals of BDS (one-hour) and GPS (two-hour) broadcast ephemerides, the daily message amounts (bottom bars) are much lower than that of Type 1697 (top bars), ranging between 60 and 173 for Type 72, and between 365 and 626 for Type 71, respectively.

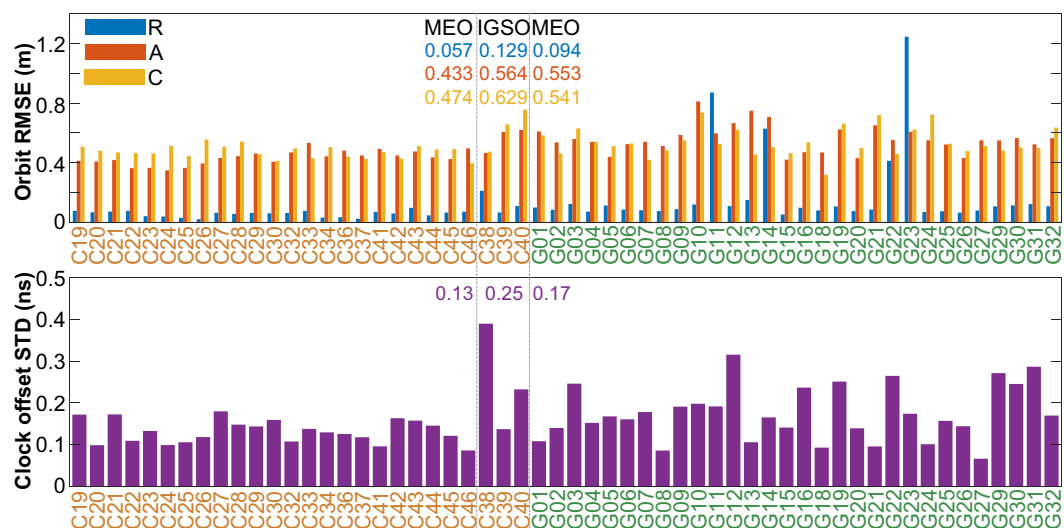


Fig. 15 Orbit RMSEs (top) and clock offset STDs (bottom) for PPP-B2b precise products during Jan 1-7, 2023.

Evaluation of Level 1. Decoding of PPP-B2b augmentation messages in the sub-folder Level_1 shows that a total of 78,246,531 PPP-B2b augmentation messages were collected in 2023, of which 78,238,323 passed the CRC check. The message amounts from C59, C60, C61, and C62 are 30,098,337, 30,088,786, 95,477, and 17,955,723, respectively. It should be noted that messages from C61 were only collected on January 19 and March 3, and no message was collected from C61 during other periods. Therefore, subsequent discussions in this study exclude C61. For messages from C62, the PPP service statuses were set to 1, indicating that the service provided by C62 was unavailable. C61 and C62 were launched on June 23, 2020 and May 17, 2023, respectively. They were in the on-orbit testing phase and not providing the same full of capability (FOC) service as C59/C60 (<https://www.bdsupport.cn/status/constellation>).

Taking the first day of each month in 2023 as examples, Fig. 11 shows the amounts of Type 1/2/3/4/63 broadcast by C59, C60, and C62. It can be observed that:

- (1) PPP-B2b augmentation messages were collected for C59 and C60 throughout all 12 months. C62 began broadcasting messages from May 24 but only broadcast Type 63 in June and July. C62 started broadcasting Type 1/2/3/4/63 simultaneously since August.
- (2) For different BDS GEO satellites, the amounts of Type 1/2/3/4/63 were close to each other after August, 2023. For example, the amount of Type 1/2/3/4/63 broadcast by C59, C60, and C62 on December 1 were 1,783/6,284/6,515/42,694/28,100, 1,731/6,155/6,386/41,559/27,279, and 1,793/6,663/6,654/42,993/27,986, respectively.

Evaluation of Level 2. This sub-section evaluates the PPP-B2b SSR corrections in the sub-folder Level_2 from both spatial and temporal perspectives. From the spatial perspective, the available satellites within PPP-B2b SSR correction are extracted, and the ground tracks of extracted satellites are calculated to monitor the coverage of BDS PPP service. Figure 12 presents the service coverage during January 1-7, 2023. It can be observed that the service coverage is within the latitude from 45.17° S to 57.33° N and longitude from 16.06° W to 98.64° W. This is due to the fact that the BDS PPP service utilizes regional tracking stations to generate PPP-B2b SSR corrections³⁷. The identical service coverage is observed from February to December, 2023 in the dataset, and also aligns with the spatial coverage in Ouyang *et al.*⁷ based on the PPP-B2b SSR corrections during June 17–21, 2022. This consistency indicates that the BDS PPP service coverage does not change from June 2022 to December 2023.

The reception latency of the PPP-B2b SSR corrections are calculated by differencing between the Message reception time and the Correction reference time (see Table 5):

$$\text{Reception Latency} = \text{Message Reception Time} - \text{Correction reference time} \quad (1)$$

Taking December 1, 2023 as an example, the amounts of satellite masks/orbit corrections/DCB corrections/clock corrections from C59, C60, and C62 are 1,783/6,284/6,515/42,694, 1,731/6,155/6,386/41,559, and 1,793/6,663/6,654/42,993, respectively. Figure 13 illustrates the reception latencies of these corrections in the form of boxplots. The top and bottom solid lines of each boxplot represent the maximum and minimum latencies, while the top and bottom of the box indicate the upper and lower quartiles, respectively. The red line inside the box represents the median latency. It can be observed that:

- (1) Reception latencies of corrections from C59 and C60 are consistent, with median latencies of 5.12 s, 24.12 s, 17.12 s, and 7.12 s for the four types of corrections, respectively.

- (2) For C62, the median latency of DCB corrections is 56.12 s, which is obviously higher than that of C59 and C60 (17.12 s). This latency difference is attributed to the fact that C62 was in on-orbit testing phase, and was not providing the same FOC PPP service as C59/C60 in 2023. Although the maximum clock offset correction latencies occasionally exceed 20 s, the median value is 7.12 s, consistent with that of C59 and C60. Median latencies for satellite mask and orbit corrections are 5.12 s and 24.12 s, which are also consistent with those of C59 and C60.

Evaluation of Level 3. This sub-section evaluates PPP-B2b precise products in the sub-folder Level_3 from two aspects: completeness rate and orbit/clock offset accuracy/precision. For PPP-B2b orbits with the 48 s sampling interval, the completeness rate is defined as:

$$\text{Completeness Rate} = \frac{\text{Epoch\#}}{T/48 \text{ s}} \times 100\% \quad (2)$$

where *Epoch #* refers to the number of epochs in the SP3 file, and *T* represents the collection duration of PPP-B2b augmentation messages. Figure 14 presents the daily completeness rate in 2023. From April 10 to 18, the receiver underwent a firmware update with the version upgraded from 601A2.p.21 to 620F1.23924.1. Excluding these nine days, the average completeness rate is 99.57% for remaining days in 2023.

PPP-B2b orbit accuracies and clock offset precisions are evaluated taking GFZ rapid products (<ftp://ftp.gfz-potsdam.de/GNSS/products/mgex>) as references. PPP-B2b orbit accuracies are assessed using RMSEs in the R/A/C directions, and clock offset precisions are evaluated using STDs of double-differenced clock offsets³⁸. Figure 15 presents the orbit RMSEs and clock offset STDs for BDS and GPS satellites during January 1-7, 2023. It can be observed that:

- (1) Average R/A/C orbit RMSEs for BDS MEO and IGSO satellites are 0.057/0.433/0.474 m and 0.129/0.564/0.629 m, respectively, with average clock offset STDs of 0.13 ns and 0.25 ns.
- (2) For GPS satellites, average R/A/C orbit RMSEs are 0.094/0.553/0.541 m, with an average clock offset STD of 0.17 ns. It is worth noting that R-direction orbit RMSEs for G11, G14, G22, and G23 are significantly larger (0.870 m, 0.628 m, 0.413 m, and 1.244 m) than that of other satellites. This is due to the usage of different satellite antenna models between PPP-B2b and GFZ orbit products¹⁹.

Data availability

The dataset is publicly available under a CC BY 4.0 open license and can be accessed via Science Data Bank at the repository: <https://doi.org/10.57760/sciencedb.cast.0000933>.

Code availability

The code for generating BDSet is available at the repository: <https://doi.org/10.57760/sciencedb.cast.0000933>. The code is developed in C++ and compiled with Microsoft Visual Studio Community 2019 under the Windows 10 Professional operating system (Version 22H2). Following successful compilation, the dataset is generated through a two-step procedure: first, the daily raw data from receiving chip and mixed broadcast ephemerides are stored in two separate directories as input preparation. Then, the compiled program is executed in command interpreter, with five input arguments: the directory of the raw data, the start and end date of data collection in YYYYDOY format, the directory of the mixed broadcast ephemeris, and the output directory for the dataset.

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References

1. Namie, H. *et al.* Performance evaluation of Centimeter-Level Augmentation Positioning L6-CLAS/MADOCa at the beginning of official operation of QZSS. *IEEE Journal of Industry Applications* **10**(1), 27–35, <https://doi.org/10.1541/ieejia.20001080> (2021).
2. Fernandez-Hernandez, I. *et al.* Galileo high accuracy service: initial definition and performance. *GPS Solutions* **26**, 65, <https://doi.org/10.1007/s10291-022-01247-x> (2022).
3. Yang, Y. *et al.* Featured services and performance of BDS-3. *Science bulletin* **66**(20), 2135–2143, <https://doi.org/10.1016/j.scib.2021.06.013> (2021).
4. CSNO. BeiDou Navigation Satellite System Signal In Space Interface Control Document: Precise Point Positioning Service Signal PPP-B2b (Beta Version). <http://www.beidou.gov.cn/xt/gfzx/201912/P020230516558050038035.pdf> (2019).
5. Tang, C. *et al.* Orbit determination, clock estimation and performance evaluation of BDS-3 PPP-B2b service. *Journal of Geodesy* **96**, 60, <https://doi.org/10.1007/s00190-022-01642-9> (2022).
6. Shi, J., Ouyang, C., Huang, Y. & Peng, W. Assessment of BDS-3 global positioning service: ephemeris, SPP, PPP, RTK, and new signal. *GPS Solutions* **24**(3), <https://doi.org/10.1007/s10291-020-00995-y> (2020).
7. Ouyang, C. *et al.* Exploring characteristics of BDS-3 PPP-B2b augmentation messages by a three-step analysis procedure. *GPS Solutions* **27**, 119, <https://doi.org/10.1007/s10291-023-01457-x> (2023).
8. Xu, Y., Zhao, P. & Wang, J. Precipitable water vapor retrieval for rainfall forecasting using BDS-3 PPP-B2b signal in the coastal region of China. *Measurement Science and Technology* **35**, 116309, <https://doi.org/10.1088/1361-6501/ad6471> (2024).
9. Yang, H., Ren, X., Liu, M. & Zhang, X. Dual-frequency to five-frequency real-time precise point positioning using new BDS-3 PPP-B2b service. *Earth, Planets and Space* **76**, 82, <https://doi.org/10.1186/s40623-024-02031-6> (2024).
10. Xu, X., Nie, Z., Wang, Z., Zhang, Y. & Dong, L. An improved BDS-3 PPP-B2b positioning approach by estimating signal in space range errors. *GPS Solutions* **27**, 110, <https://doi.org/10.1007/s10291-023-01455-z> (2023).
11. Xu, Z. *et al.* Temporal Characteristics Based Outlier Detection and Prediction Methods for PPP-B2b Orbit and Clock Corrections. *Remote Sensing* **16**, 2337, <https://doi.org/10.3390/rs16132337> (2024).

12. Tao, J., Zhang, G., Chen, G., Jiang, Y. & Zhao, Q. Real-time estimation of multi-frequency phase observable-specific bias for the BDS3 PPP-B2b service. *GPS Solutions* **29**, 19, <https://doi.org/10.1007/s10291-024-01776-7> (2025).
13. Zhou, P. *et al.* Real-Time Precise Zenith Tropospheric Delay Estimation with BDS PPP-B2b, Galileo HAS, and QZSS MADOCA-PPP Services. *IEEE Transactions on Geoscience and Remote Sensing* **62**, 5802011, <https://doi.org/10.1109/tgrs.2024.3443884> (2024).
14. Wang, J. *et al.* Multi-frequency smartphone positioning performance evaluation: insights into A-GNSS PPP-B2b services and beyond. *Satellite Navigation* **5**, 25, <https://doi.org/10.1186/s43020-024-00146-5> (2024).
15. Ouyang, C. *et al.* User Implementation and Assessment of BDS-3 Precise Point Positioning Augmentation Service with No Economic Cost. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences* **X-1/W1-2023**, 679–686, <https://doi.org/10.5194/isprs-annals-X-1-W1-2023-679-2023> (2023).
16. Tao, J. *et al.* Initial assessment of the BDS-3 PPP-B2b RTS compared with the CNES RTS. *GPS Solutions* **25**, 131, <https://doi.org/10.1007/s10291-021-01168-1> (2021).
17. Xu, Y., Yang, Y. & Li, J. Performance evaluation of BDS-3 PPP-B2b precise point positioning service. *GPS Solutions* **25**, 142, <https://doi.org/10.1007/s10291-021-01175-2> (2021).
18. Ren, Z. *et al.* Performance assessment of real-time precise point positioning using BDS PPP-B2b service signal. *Advance in Space Research* **68**(8), 3242–3254, <https://doi.org/10.1016/j.asr.2021.06.006> (2021).
19. Sun, S., Wang, M., Liu, C., Meng, X. & Ji, R. Long-term performance analysis of BDS-3 precise point positioning (PPP-B2b) service. *GPS Solutions* **27**, 69, <https://doi.org/10.1007/s10291-023-01409-5> (2023).
20. Dai, W., Qi, Q., Pan, L. & Cai, C. Real-time uncombined PPP using BDS-3 PPP-B2b products with different multi-frequency integrations and refined stochastic model. *Advances in Space Research* **71**, 5569–5579, <https://doi.org/10.1016/j.asr.2024.08.051> (2024).
21. Shi, J. *et al.* A data-driven PPP-B2b satellite clock offset prediction method considering the cross-arc statistical similarity. *GPS Solutions* **29**, 98, <https://doi.org/10.1007/s10291-025-01845-5> (2025).
22. Xu, X., Nie, Z., Wang, Z., Wang, B. & Du, Q. Performance Assessment of BDS-3 PPP-B2b/INS Loosely Coupled Integration. *Remote Sensing* **14**, 2957, <https://doi.org/10.3390/rs14132957> (2022).
23. Shang, R. *et al.* Lightweight seamless lane-level positioning with the integration of BDS-3 PPP-B2b and VINS: a case study using smartphone. *GPS Solutions* **28**, 185, <https://doi.org/10.1007/s10291-024-01730-7> (2024).
24. Guo, W., Li, L., Zhao, D., Zhu, F. & Lai, L. Enhancing BDS-3 PPP-B2b real-time positioning performance by tightly integrating MEMS IMU and LiDAR in GNSS-Degraded environment. *Measurement* **242**, 116015, <https://doi.org/10.1016/j.measurement.2024.116015> (2025).
25. Shi, J., Xu, C., Guo, J. & Gao, Y. Real-Time GPS Precise Point Positioning-Based Precipitable Water Vapor Estimation for Rainfall Monitoring and Forecasting. *IEEE Transactions on Geoscience and Remote Sensing* **53**(6), 3452–3459, <https://doi.org/10.1109/tgrs.2014.2377041> (2015).
26. Li, X. *et al.* Multi-GNSS Meteorology: Real-Time Retrieving of Atmospheric Water Vapor from BeiDou, Galileo, GLONASS, and GPS Observations. *IEEE Transactions on Geoscience and Remote Sensing* **53**(12), 6385–6393, <https://doi.org/10.1109/tgrs.2015.2438395> (2015).
27. Wang, X. *et al.* An investigation of PPP-B2b coverage and its performance in ZTD estimation and positioning in different regions. *Survey Review* **1**–15. <https://doi.org/10.1080/00396265.2024.2401665> (2024).
28. Cao, Y. *et al.* Performance of Ground-Based Global Navigation Satellite System Precipitable Water Vapor Retrieval in Beijing with the BeiDou B2b Service. *Remote Sensing* **16**(16), 2902, <https://doi.org/10.3390/rs16162902> (2024).
29. Hou, C., Shi, J., Ouyang, C., Guo, J. & Zou, J. A dual-base station constraint method to improve deformation monitoring precision consistency in strip regions. *Satellite Navigation* **5**(1), 26, <https://doi.org/10.1186/s43020-024-00148-3> (2024).
30. Geng, T. *et al.* Real-time ocean precise point positioning with BDS-3 service signal PPP-B2b. *Measurement* **203**, 111911, <https://doi.org/10.1016/j.measurement.2022.111911> (2022).
31. Shi, J., Ouyang, C., Peng, W., Huang, Y. & Xu, C. A Simplified BDS Broadcast Ephemeris and State Space Representative (SSR) Matching Method for BDS-only Real-time Precise Point Positioning (PPP). *IEEE Access* **7**, 125205–125213, <https://doi.org/10.1109/ACCESS.2019.2938252> (2019).
32. CSNO. BeiDou Navigation Satellite System Signal In Space Interface Control Document: Precise Point Positioning Service Signal PPP-B2b (Version 1.0). <http://www.beidou.gov.cn/xt/gfzx/202008/P020230516574071340728.pdf> (2020).
33. Ouyang, C. *et al.* BDS-3: a first-ever BDS satellite-based PPP augmentation dataset since 2023. V2. *Science Data Bank* <https://doi.org/10.57760/sciencedb.cast.00009> (2026).
34. Hilla, S. The Extended Standard Product 3 Orbit Format (sp3 version d). <https://files.igs.org/pub/data/format/sp3d.pdf> (2016).
35. Ray, J. RINEX Extensions to Handle Clock Information (clock RINEX 3.04). https://files.igs.org/pub/data/format/rinex_clock304.txt (2017).
36. Schaer, S. SINEX BIAS-Solution (Software/technique) Independent EXchange Format for GNSS Biases Version 1.00. https://files.igs.org/pub/data/format/sinex_bias_100.pdf (2018).
37. Yang, Y. *et al.* Principle and performance of BDSBAS and PPP-B2b of BDS-3. *Satellite Navigation* **3**(1), 5, <https://doi.org/10.1186/s43020-022-00066-2> (2022).
38. Ouyang, C., Shi, J., Huang, Y., Guo, J. & Xu, C. Evaluation of BDS-2 real-time orbit and clock corrections from four IGS analysis centers. *Measurement* **168**, 108441, <https://doi.org/10.1016/j.measurement.2020.108441> (2021).
39. Ramabadran, T. V. & Gaitonde, S. S. A tutorial on CRC computations. *IEEE Micro* **8**(4), 62–75, <https://doi.org/10.1109/40.7773> (1988).

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Author contributions

Chenhao Ouyang: Data Curation, Methodology, Software, Writing - Original Draft. Junbo Shi: Conceptualization, Writing - Review & Editing, Supervision. Cheng Hou and Wenjie Peng: Data Curation, Formal analysis. Jiming Guo and Yibin Yao: Methodology, Writing - Review & Editing.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to J.S.

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